Architecture variants

Frontier modellarning architecture lar; farq qiladi; Hammasi har xil javob beradi savollarga.

Nima uchun qo'llaniladi: Rivojlantirilk

- -> Faster.
- -> Better.
- -> Less memory
- -> Longer Contex

Trade off

Balans kerak

Attention turlari:

MHA (Multi Head Attention) – eski modellar use

MQA (Multi Query Attention)

GQA (Group Query Attention)

GPT 2 multi Head Attention

d_model => 768.

12 -> attention heads

768 / 12 -> 64 – Dimensions per head.

768 -> 2304.

Query (Q) 768

Key (K) 768

Value (V) 768.

Q, K, V

↓

Head1 64

⋮

Head12 64.

(head 1) 64

(head 2) 64

(head 3) 64

⋮

(head 12) 64

768

Concatenate.

768

MHA → GPT2 da ishlatilgan. Eski model. Hozirgi yangi modellar (GQA) ishlatadi (Balansni ushlaydi)

GPT2: MHA, Absolute embedding, GELU, LayerNorm, BPE (~50k tokens), Good language modeling.

New models: (GQA, MQA) RoPE (Rotary positional Em), SwiGLU, RMSNorm, Larger BPE,
- Faster inference, longer context windows, lower memory overhead, better output quality.

Residual

Malumotlar yo'qolishini oldini oladigan operation.

① $x_1 = x + \text{Attention}(\text{LayerNorm}(x))$

② $x_2 = x_1 + \text{FFN}(\text{LayerNorm}(x))$.

Ollama

→ AI modellarning playstore desak bo'ladi.
→ local LLM lar mavjud.